

Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions



Dataset Overview

3,900

Purchase Records

Transactional data analyzed

18

Data Fields

Customer and purchase details

37

Missing Values

Review ratings imputed

Key features include customer demographics, purchase details, and shopping behavior metrics.

Data Preparation Process

01

Data Loading

Imported dataset using pandas for analysis

02

Exploration

Checked structure with `info()` and `describe()`

03

Cleaning

Handled missing values and standardized columns

04

Feature Engineering

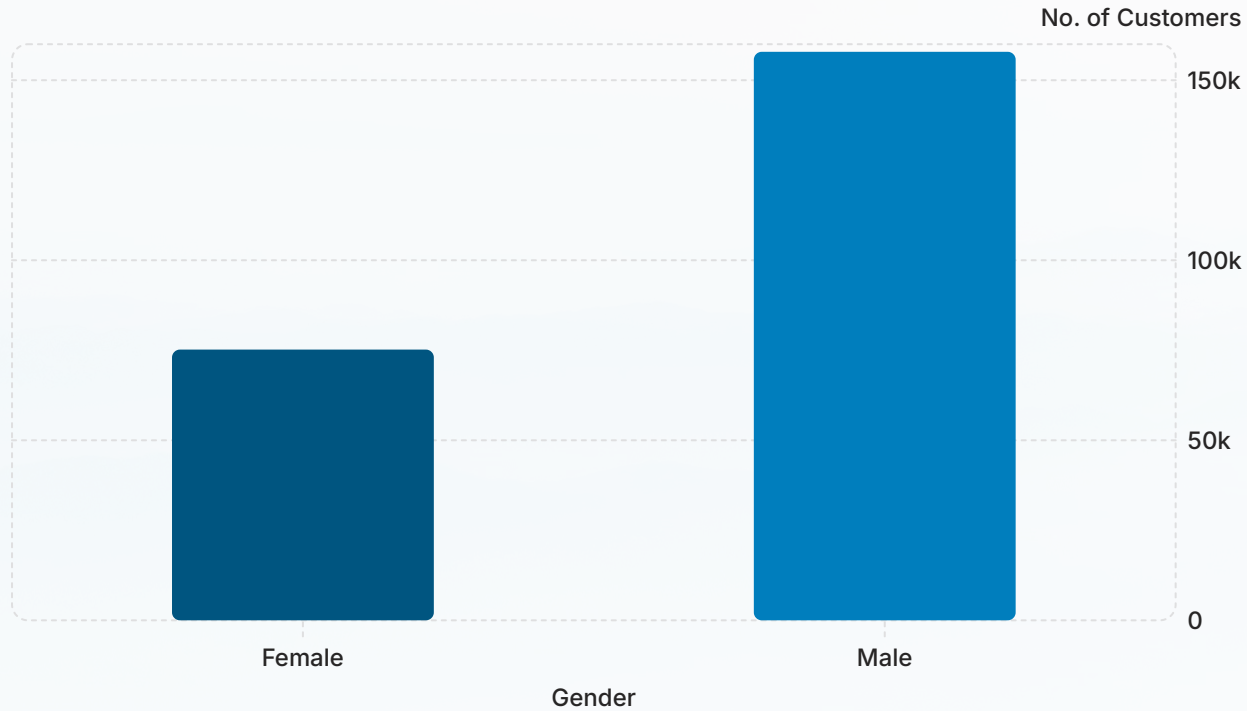
Created age groups and purchase frequency metrics

05

Database Integration

Loaded cleaned data into PostgreSQL

Revenue by Gender



Key Finding

Male customers generated **\$157,890** in revenue, significantly outpacing female customers at **\$75,191**.

Top Products by Rating

1

Gloves

3.86 average rating

2

Sandals

3.84 average rating

3

Boots

3.82 average rating

4

Hat

3.80 average rating

5

Skirt

3.78 average rating

These products received the highest customer satisfaction scores.



Shipping & Subscription Impact

Express Shipping

\$60.48 average purchase vs.
\$58.46 for standard

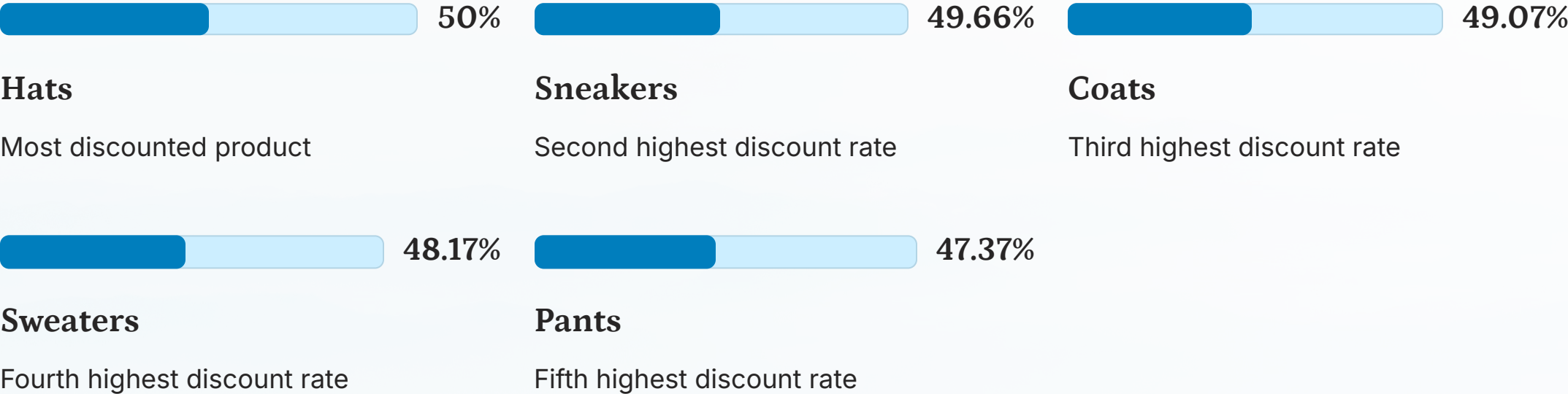
Non-Subscribers

\$59.87 avg spend, 2,847
customers, \$170,436 total
revenue

Subscribers

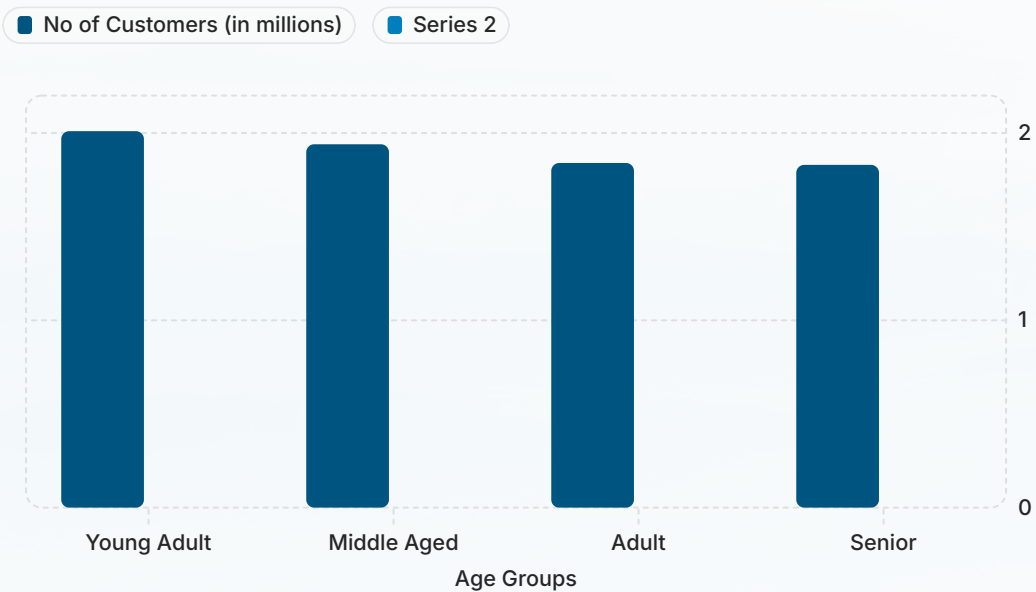
\$59.49 avg spend, 1,053 customers, \$62,645 total revenue

Discount Strategy Insights



839 customers used discounts while spending above average purchase amounts.

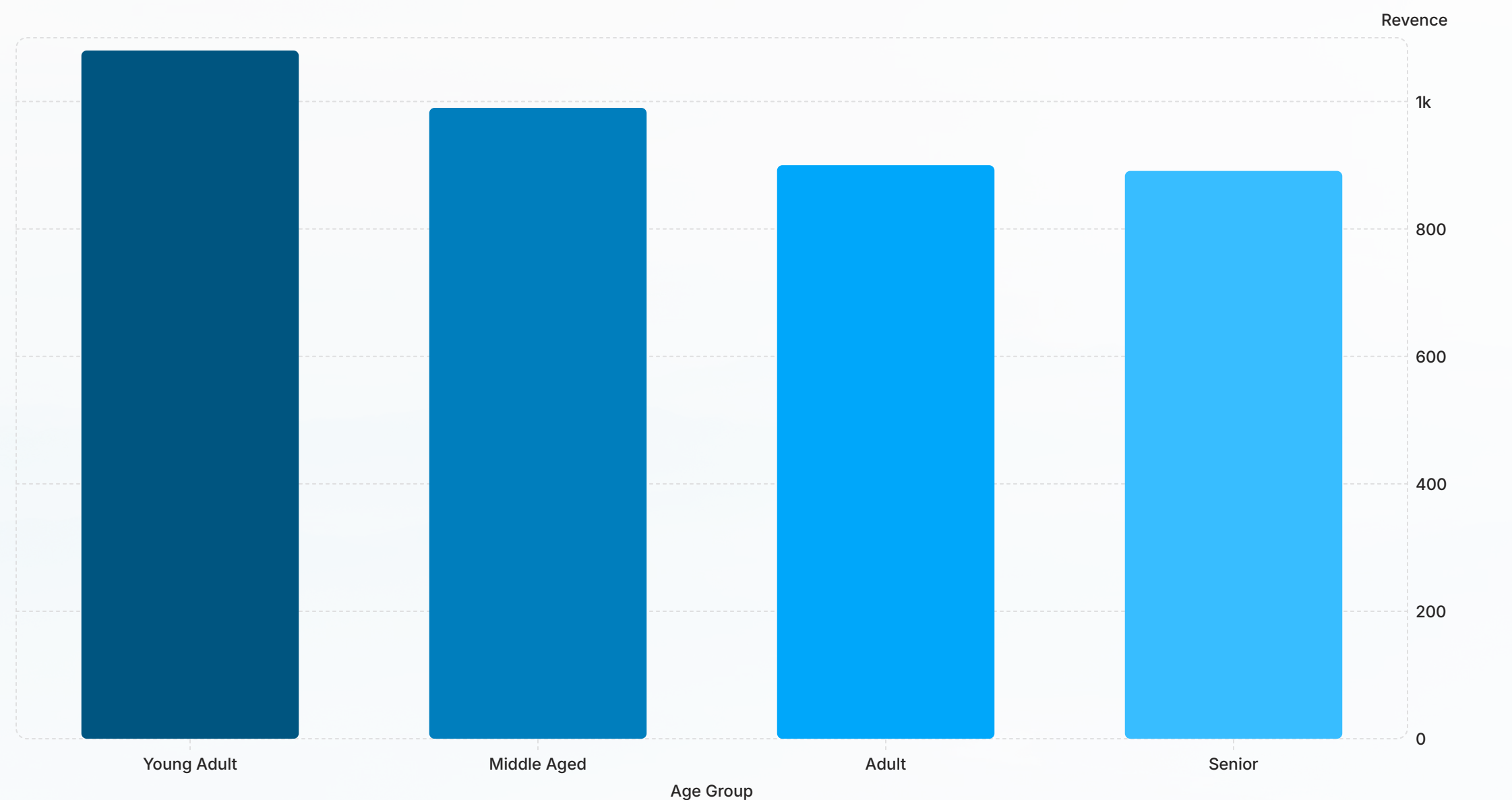
Customer Segmentation



Segment Distribution

Majority of customers (3,116) are loyal buyers, while 701 are returning and 83 are new customers.

Revenue by Age Group



Young adults generated the highest revenue at **\$62,143**, followed by middle-aged customers at **\$59,197**.

Strategic Recommendations



Boost Subscriptions

Promote exclusive benefits for subscribers



Customer Loyalty Programs

Reward repeat buyers to move them into loyal segment



Review Discount Policy

Balance sales boosts with margin control



Targeted Marketing

Focus on high-revenue age groups and express-shipping users

